

Legenda

- $t1 \in \text{typeSet} \rightarrow$ per dire che "t1" appartiene all'insieme dei tipi consentiti per i Set
- $\text{sameType}(a,b) \rightarrow$ per dire che "a" e "b" sono dello stesso tipo
- $"=" \rightarrow$ restituisce "true" sse i due valori sono uguali, "false" altrimenti
- $"\cup", "\cap", "-" \rightarrow$ operatori insiemistici di unione, intersezione e differenza

Regole Operazionali

- Empty Set (t)

$$\frac{t \in \text{typeSet}}{\text{Env} \triangleright \text{EmptySet}(i) \rightarrow \text{SetVal}(\text{Empty}(t))}$$

- Singleton (e, t)

$$\frac{t \in \text{typeSet}, \text{Env} \triangleright v \rightarrow e, \text{sameType}(t, e)}{\text{Env} \triangleright \text{Singleton}(v, i) \rightarrow \text{SetVal}(\text{Set}([e], t))}$$

- Of (t, l)

$$\frac{t \in \text{typeSet}, \text{Env} \triangleright l \rightarrow l1, (\forall x \in l1. \text{sameType}(x, t1))}{\text{Env} \triangleright \text{Of}(t, l) \rightarrow \text{SetVal}(\text{Set}(l1, t1))}$$

- Insert (e, s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright e \rightarrow v1, \text{sameType}(t, v1)}{\text{Env} \triangleright \text{Insert}(e, s) \rightarrow \text{SetVal}(\text{Set}(l1 \cup v1, t))}$$

- Remove (e, s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright e \rightarrow v1, \text{sameType}(t, v1)}{\text{Env} \triangleright \text{Remove}(e, s) \rightarrow \text{SetVal}(\text{Set}(l1 - v1, t))}$$

- Belongs to (e, s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright s \rightarrow v1, \text{sameType}(t, v1)}{\text{Env} \triangleright \text{Belongs to}(e, s) \rightarrow ((l1 \cap v1) = v1)}$$

- Is empty (s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{Set}(l1, t)}{\text{Env} \triangleright \text{IsEmpty}(s) \rightarrow (s = \text{SetVal}(\text{Set}([], t)) \vee s = \text{SetVal}(\text{Empty}(t)))}$$

- Get min (s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t)), (\exists x \in l1. (\forall y \in l1. x < y))}{\text{Env} \triangleright \text{Get min}(e) \rightarrow x}$$

- Get max (s)

$$\frac{\text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t)), (\exists x \in l1. (\forall y \in l1. x > y))}{\text{Env} \triangleright \text{Get max}(s) \rightarrow x}$$

- Is subset (s1, s2)

$$\frac{\text{Env} \triangleright s1 \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright s2 \rightarrow \text{SetVal}(\text{Set}(l1, t))}{\text{Env} \triangleright \text{Is subset}(s1, s2) \rightarrow ((l1 \cap l2) = l2)}$$

- Union (s1, s2)

$$\frac{\text{Env} \triangleright s1 \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright s2 \rightarrow \text{SetVal}(\text{Set}(l2, t))}{\text{Env} \triangleright \text{Union}(s1, s2) \rightarrow \text{SetVal}(\text{Set}(l1 \cup l2, t))}$$

- Intersection (s1, s2)

$$\frac{\text{Env} \triangleright e1 \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright e2 \rightarrow \text{SetVal}(\text{Set}(l2, t))}{\text{Env} \triangleright \text{Intersection}(s1, s2) \rightarrow \text{SetVal}(\text{Set}(l1 \cap l2, t))}$$

- Difference (s1, s2)

$$\frac{\text{Env} \triangleright s1 \rightarrow \text{SetVal}(\text{Set}(l1, t)), \text{Env} \triangleright s2 \rightarrow \text{SetVal}(\text{Set}(l2, t))}{\text{Env} \triangleright \text{Difference}(s1, s2) \rightarrow \text{SetVal}(\text{Set}(l1 - l2, t))}$$

- For all (p, s)

$$\frac{\text{Env} \triangleright p \rightarrow \text{FunVal}(\text{arg}, \text{fbody}, \text{fDecEnv}), \text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t))}{(\forall x \in l1. \text{fDecEnv}[l \setminus x] \triangleright \text{fbody} \rightarrow \text{sameType}(v, \text{Bool})), (\forall y. y = \text{Bool}(\text{true})) \rightarrow \text{ret}} \text{Env} \triangleright \text{Exists}(p, s) \rightarrow \text{ret}$$

- Exists (p, s)

$$\frac{\text{Env} \triangleright p \rightarrow \text{FunVal}(\text{arg}, \text{fbody}, \text{fDecEnv}), \text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t))}{(\forall x \in l1. \text{fDecEnv}[l \setminus x] \triangleright \text{fbody} \rightarrow \text{sameType}(v, \text{Bool})), (\exists y. y = \text{Bool}(\text{true})) \rightarrow \text{ret}} \text{Env} \triangleright \text{Exists}(p, s) \rightarrow \text{ret}$$

- Filter (p, s)

$$\frac{\text{Env} \triangleright p \rightarrow \text{FunVal}(\text{arg}, \text{fbody}, \text{fDecEnv}), \text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t))}{(\forall x \in l1. \text{fDecEnv}[l \setminus x] \triangleright \text{fbody} \rightarrow \text{sameType}(v, \text{Bool})), (\forall y \in l1. (z = \text{Bool}(\text{true})) \rightarrow y) \rightarrow \text{ret}} \text{Env} \triangleright \text{Filter}(p, s) \rightarrow \text{ret}$$

- Map (p, s)

$$\frac{\text{Env} \triangleright p \rightarrow \text{FunVal}(\text{arg}, \text{fbody}, \text{fDecEnv}), \text{Env} \triangleright s \rightarrow \text{SetVal}(\text{Set}(l1, t))}{(\forall x \in l1. \text{fDecEnv}[l \setminus x] \triangleright \text{fbody} \rightarrow y) \rightarrow \text{ret}} \text{Env} \triangleright \text{Map}(p, s) \rightarrow \text{ret}$$